Lesson Notes

MODULE 1 | LESSON 4

SURVEYING THE FINANCIAL INDUSTRY

Reading Time	60 minutes
Prior Knowledge	Bonds, Interest Rates, Credit Risk
Keywords	Central Bank, Monetary Policy, Investment Banks, Market Risk, Reinvestment Risk, Default Risk

In the previous module, we looked at bonds as another security in which lenders provide funding to borrowers. We also saw that buy-side and sell-side institutions comprise the financial industry. In this lesson, we examine different types of banks. We'll also look at different types of risks associated with bonds.

1. Central Banks

While we have discussed distinct kinds of banks, there is one type of bank we have yet to discuss: the central bank. Central banks are financial institutions whose purpose is to maintain stability and transparency for countries' economies. Central banks help to keep inflation in check, provide maximum employment opportunities, and enforce healthy monetary policy for a stable economy. Please refer to the required reading by Fotia and Bindseil, Chapter 1 and Chapter 3.

The responsibilities of central banks are complex and tend to vary by country. However, one responsibility of central banks is to help set operational targets. As you review the required reading, note that the ultimate targets are the inflation rate and the foreign exchange rate. Central banks can set interest rates at which banks borrow and lend from each other. By doing so, they can help ensure that there are sufficient funds that can be provided between banks for them to conduct their business.

2. Investment Banks: Functions, Maturities, and Products

Different from central banks are investment banks. Investment banks have distinct functions:

1. They underwrite securities.
2. They help with IPO issuance.

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3. They engage in sales and trading with investors.
4. They provide research for clients.
5. They bundle and sell securitized products.
6. They perform mergers and acquisitions.

Investment banks will also help **underwrite new businesses**. One thing that they must do is provide credit to businesses. Once again, in our module, we see credit risk. Investment banks continually monitor the credit risks of their clients, of their investments, and of each other. If you are a business, then you have a measure that will constitute your credit score. Investment banks will look at this credit score to determine the amounts and even rates at which you are charged for the use of funds.

Another purpose of banks is to **provide credit to individuals.** Individuals need a quantitative measurement of their credit. For example, in the United States, there is something known as a FICO score. FICO is an acronym for Fair Isaac's Company. This company developed a quantitative model that considers multiple factors in determining the credit worthiness of an individual and distills them down to a single number. That number is on a spectrum ranging from 300 to 850. Based on someone's FICO score, a person has credit worthiness that is made readily available to lenders: credit card lenders, mortgage lenders, property owners, and other financial institutions. Imagine the challenging work required if each prospective stakeholder in those examples would have to conduct their own credit risk assessment of an individual. These credit scores help recognize and reward individuals who build and maintain good credit by allowing them the use of credit at rates that reflect their creditworthiness.

Banks can also provide an array of fixed-income products ranging from very short-term loans and certificates of deposit to intermediate and advanced products:

- Short-term: typically less than 1 year.
- Intermediate-term: typically 1 to 10 years.
- Long-term: typically more than 10 years.

There are distinct types of products, including:

- Commercial paper
- Certificates of deposits
- Treasury bills
- Treasury notes
- Treasury bonds

3. Commercial Banks

Most of us are perhaps more familiar with commercial banks. We tend to think of them as places to deposit money, get a credit card, and get loans and/or mortgages. Commercial banks were highly regulated and their daily activities were integrated inside a wider financial system. Specifically, commercial banks were mostly concerned with attracting deposits from the public and lending money to individuals and businesses. In the U.S., the *Glass-Steagall Act* of 1933 made sure that there was a

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clear distinction between commercial banks and investment banks. That had important consequences for big banks like JP Morgan, which had to cut their services and lost several sources of income. Banks tried to sidestep the Glass-Steagall Act by using holding companies to avoid the restrictions on bank branching or having multiple offices. The *Bank Holding Company Act* of 1956 gave the Federal Reserve more oversight of this. In 1999, the Glass-Steagall Act was repealed, and today, the distinction between investment banks and commercial banks, for practical purposes, has little or no meaning--so much so that currently JPMorgan Chase Bank is among the largest commercial banks in the U.S. and also among the leaders in underwriting IPOs.

Many believe that the repeal of the Glass-Steagall Act allowed a consolidation (via mergers and acquisitions) of investment and commercial banks. Such consolidation in turn led to financial institutions that heavily invested in mortgage-backed securities, which, as we know, led to the big subprime mortgage crisis and its ultimate implosion in 2008-9. In fact, before the repeal of the Glass-Steagall Act, investment banks (the banks that generated the subprime mortgage derivatives) would not have been able to find themselves involved in mortgage lending as well (typically done by commercial banks). However, there are also those who counter that this is not true because big players like Lehman Brothers, Bear Stearns, and Goldman Sachs were big players in the meltdown of subprime mortgage investments, yet they were never commercial banks.

It has been theorized that the subprime mortgage meltdown was actually caused by the Department of Housing and Urban Development, which required Fannie Mae and Freddie Mac to purchase more affordable mortgages to encourage lenders to make loans to low-income and minority borrowers. Commercial banks, to meet HUD's objectives, began to give out mortgage loans to people who had no money for down payments or accepted unemployment benefits as a valid source of income. However, many of these lenders were not banks. Countrywide Financial, one of the main culprits, was not a commercial bank strictly speaking as it did not accept deposits. So, again, it is hard to argue that the cause of the subprime mortgage crisis should be attributed to the lack of distinction between commercial and investment banking.

Countrywide Financial is an ideal example to use to introduce some of the arguments for our next section: the existence of risks. Countrywide, under CEO Angelo Mozilo, pumped out complex mortgages to borrowers who did not understand them and could not afford them. Borrowers simply did not understand the risks involved with their mortgages, and Countrywide did not care much about what happened after the borrowers signed their mortgages because it packaged them up and sent them to Wall Street to sell as securitized products.

4. Types of Risks

Each financial product has its own set of risks. Some of them we already considered in early lessons while others are new. In any case, we can use this as a review. Consider, for example, that you have a choice between depositing money in a savings account, such as a certificate of deposit, or purchasing a bond. As such, you want to know what the risks are. Let's consider four different risks.

4.1 Market Risk

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Market risk is the risk due to price fluctuation, which makes the value of assets go down. If you have a savings account, then you are immune to market risk because you have a specified interest rate. However, if you own a bond, the prices do change. If you bought a bond and then try to sell it, you might find that you have to sell at a lower price. As a result, you would lose money due to market fluctuation. One safe way to avoid market risk in a bond is simply to hold the bond to maturity. If you hold a bond to maturity, then the bond price can change. So long as there is no default, you are immune to price fluctuations because you never sell the bond.

4.2 Reinvestment Risk

If you decide to go with the bond, then periodically, you will receive coupon interest. You may decide not to spend that coupon interest but rather to reinvest it. What will interest rates be like when you do sell? If interest rates go up, then that means you will be able to reinvest at a higher rate of interest, which is good. However, if interest rates go down, then you would reinvest at a lower rate of interest. The risk is there. **Reinvestment risk** is the risk that rates are lower when you have access to new investment funds. Reinvestment risk exists for CDs because, upon maturity, as you seek to renew a certificate of deposit, the rate may be lower than what you originally had.

4.3 Default Risk

Default risk is the risk that your bond issue fails to pay you back the stated interest and/or the principal. As we have seen, the certificate of deposit is insured by our trustworthy depository insurance corporation, so default risk is a non-issue. If you were the holder of a risk-free bond, then likewise, you need not worry about the default risk of those securities. However, if you are the buyer and/or holder of a risky bond, then you are certainly subject to the default risk of the bond issuer. If the bond issuer defaults, then the bond will be worth less. Even in the absence of a default, the fact that the bond issuer's credit rating declines causes the bond to be worth less as the bond is perceived as riskier than before.

Again, let's look at the difference between holding the bond to maturity versus trading the bond. If you simply hold the bond to maturity, then you run the risk that there may be a default. This means that you lose the interest and principal. If you only hold the bond for a brief period of time, then you are still subject to the credit risk. The decline in credit will cause a market drop, so although it is not default risk specifically, it is credit risk in general. Default risk is just an extreme form of credit risk. Any lower credit rating results in a price drop of the bond. As a result, when you go to sell the bond, you will undoubtedly get less due to that higher credit risk. Default risk is just one type--the main type--of credit risk. Credit risk really refers to the degradation of credit worthiness. At a small scale, this results in a simple lowering of the credit score. At an extreme scale, this is actual default.

4.4 Inflation Risk

Inflation risk is the risk that the money you receive is simply worth less because inflation was higher than expected. For example, suppose either the certificate of deposit or the bond earned you 3% interest over the year. But what if inflation were 8% that year? Indeed, your real rate of return would be -5 percent. That is, had you gone to the marketplace, you could have purchased goods and services that now, by delaying a year, cost you more money; even though you have 3% more, those goods and services cost 8% more. In fact, people often find themselves in this situation in high-inflation

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environments. This is where central banks can help. When central banks find inflation to be high, they look to perform monetary policy, like raising interest rates, that will help to keep inflation under control.

4.5 Credit Risk Revisited

Lenders must consider several key types of risk when they originate loans:

- fraud risk;
- default risk;
- credit spread risk; and
- concentration risk

Altogether, these 4 risks are referred to as credit risk for banks or lenders in general.

Fraud risk has to do with the verification of the person or company that applies for a loan in order to distinguish legitimate applications from those that are fraudulent. Lenders need to check if the borrower is serious about repaying their debt. For instance, a client may ask to take out a loan in their name without any intention to repay it. Instead, the client may use the loan for an exotic vacation. Some would say that Countrywide Financial did not care about this at all as they often extended credit to individuals who could not really afford it. The following optional reading Owner-Occupancy Fraud and Mortgage Performance could be of interest for some students as it highlights instances of fraud.

Default risk was explained above in detail. It is worth noting that default risk is connected to **counterparty risk**. In some complicated transactions, the parties involved promise to exchange certain cash flows or exchange cash for assets. If one of the parties does not fulfill its obligations under the contract, it could lead to financial disruption for the other party. In turn, the latter party could default on its obligation toward a different third party, and so on. Therefore, lenders need to ascertain whether all parties involved can be counted on to fulfill their contractual obligations and/or need to decide how much credit to extend to the parties, if any.

Credit spread risk. Commercial bonds and risk-free government bonds or Treasury bills have a credit spread that depends on perceived levels of risk. If the spread increases, the credit risk for the borrower increases. The market value of the assets could decrease, which would result in losses for lenders and/or investors.

Concentration risk. This type of risk should be intuitive. When a large portion of a bank's portfolio is concentrated on a single sector or asset class (think of Countrywide Financial and the fact that almost their entire balance sheet was tied to the real estate industry) and when adverse events affect that sector or asset class, this concentration can cause substantial losses. Some argue that concentration could also contribute to systemic risk as the failure in one sector could cause failures in others, resulting in a domino effect.

5. Types of Marketplaces

We have already discussed different types of maturities, products, and risks, so now, we'll close this lesson by showing that there are distinct types of marketplaces as well. Broadly speaking, there are two

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kinds. The first type of marketplace is an **exchange**. The second is what is called **over the counter**, abbreviated as **OTC**.

Exchanges have the following characteristics:

- They offer a variety of benefits that OTC markets do not have. The main advantages of exchanges are efficiency, transparency, equitability, and credit risk mitigation.
- They primarily provide a centralized marketplace, making trading efficient. They have very well-defined rules for handling the processing and clearing of trades.
- They provide transparency because all market participants have access to the order screens that contain the bid sizes and amounts, the offer sizes and amounts, and a record of the previous trades that took place.
- The transparency and availability of data makes trading fairer for market participants. Of course, big institutional investors may have bigger infrastructure with fast computers and smart algorithms to detect and process orders more efficiently than smaller individuals or institutions do.
- They solve the problem of credit risk. When you perform a transaction on an exchange, all the credit risk is eliminated. How? The credit risk is taken by the exchange itself. In other words, suppose someone named Mamadou borrowed a security from you in order to sell it short. Soon after Mamadou sells it, the price jumps up! Now, you have more credit risk because it just became more difficult for Mamadou to cover that short. The exchange can require Mamadou to post additional financing. Financing was one of the two challenges in this module (with the other being credit risk). The exchange continually monitors the value of that security and requires Mamadou to post additional funds, even as often as intraday. In doing so, you as the lender of the security would be assured that, even with price increases, you will be able to receive the security back because Mamadou cannot run out of money to cover it. The exchange enforces this by mandating intraday requirements such that Mamadou must post cash in an account to make sure he has sufficient funds available to cover that short. If Mamadou were not able to do so, the exchange could force the sale to immediately cover the security and return it to the lender. So, the exchange's monitoring and enforcement of margin helps to ensure that financing remains creditworthy.

Ideally, all securities would be exchange traded for the advantages of equitability, transparency, and the minimization of credit risk. When we turn to OTC markets, we see that trading occurs directly between counterparties. Yes, there may be brokers in between, but the brokers do not take credit risk. The brokers are there to facilitate and manage the transaction, not to assume the credit risk of either side. Likewise, the transparency is more difficult because OTC markets decentralize the process. For example, stocks that trade on a stock exchange only trade in that one place. However, bonds may trade OTC for which there are different OTC markets. That means that selected OTC markets can have more market share than others. As a result, there could be slightly different prices from time to time between markets. Therefore, it is much harder to assure that you have received a fair price.

6. Conclusion

We started off this lesson by introducing central banks. For now, we want to emphasize that banks help to address the challenge we've seen in this module: credit risk. Credit risk appears in several places; in

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this module, we discussed the challenge of financing. We concluded by showing that the type of financial trading, namely exchanges, can also help mitigate credit risk. However, there are still many products that trade over the counter, so credit risk continues to be an ongoing concern for many financial transactions.

In the next lesson, we will switch asset classes by moving from fixed income to equities and cryptocurrencies. Those assets have much more volatility than fixed income does. Just as credit risk is a major challenge to fixed income, volatility will be a significant consideration for equities and cryptocurrencies.

References

- Fotia, Alessio, and Ulrich Bindseil. “Introduction to Central Banking.” *Conventional Monetary Policy*, Springer, Cham, 2021, pp. 1–9.
- Fotia, Alessio, and Ulrich Bindseil. “Economic Accounts and Financial Systems.” *Conventional Monetary Policy*, Springer, Cham, 2021, pp. 21–51.

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